

Journalist Risk and threshold

Scenario: A 20-patient dataset has been generalized. After grouping, 3 records have group size $k=1$ (unique), 10 have $k=2$, and 7 have $k=3$. Compute Journalist Risk. Does this exceed the threshold?

1. Compute Journalist Risk

- **Definition:** Journalist Risk assumes an attacker (the journalist) has external information about specific individuals and tries to re-identify them by looking for unique records in the dataset.
- **Formula:** It is typically computed as the proportion of records that are unique ($k=1$) in the released dataset.
- **Calculation:**
 - Total Patients: 20
 - Unique Records ($k=1$): 3
 - **Journalist Risk** = $\frac{\text{Unique Records}}{\text{Total Records}} = \frac{3}{20} = \mathbf{0.15}$ (or 15%). [1]

2. Does this exceed the threshold?

- **Yes**, a 15% risk exceeds commonly accepted thresholds.
- **Threshold:** Health Canada and many health data regulators recommend a maximum re-identification risk threshold of **0.09 (or 9%)**.
- **Conclusion:** The calculated 15% risk is higher than 9%, meaning the dataset is likely **not sufficiently de-identified** for public release under standard guidelines